AI 2035

Plan P & Plan C : The Sovereign Divergence

Autonomous Cyber Threats to Critical Infrastructure: A Calibrated Net Assessment of Offensive AI Scaling, Geopolitical Branching, and Taiwan Supply-Chain Dependencies

A scenario on the offensive scaling of autonomous cyber capability
critical infrastructure & semiconductor geography · 2035-ai.com

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FOREWORD

Most debates about AI fixate on Artificial General Intelligence (AGI) or Artificial Superintelligence (ASI): hypothetical models out-thinking human experts across every intellectual domain.[1] In *AI 2035*, I explore an earlier, more neglected, but immediately dangerous threshold: the point where autonomous AI agents scale offensive cyber operations across enterprise networks and industrial control systems (ICS). This pathway does not require human-level AGI or superhuman ASI. The decisive transition occurs when a single operator can spin up thousands of concurrent agents capable of independently discovering zero-days, writing multi-step exploit chains, moving laterally, and manipulating operational technology protocols.[2]

The capability levels in this scenario (spanning Level C0 in 2024 through Level C7 at the 2031 strategic crossroads, branching into Plan P and Plan C through 2035) are strictly empirical capability denominations, not calendar-driven assumptions. They denote crossing discrete operational milestones, from prompt-assisted research (C0) to autonomous exploit discovery (C2), live network evasion (C3), physical industrial protocol manipulation (C4), mass cloud concurrency (C6), and cross-sector cascading failure (C7). The progress of autonomous offensive capability accelerates through three compounding dynamics: labor compression that collapses the skill floor (automating hundreds of engineering hours into seconds of compute and allowing novice operators to wield nation-state tradecraft), automated fuzzing loops that eliminate exploit-writing latency, and cloud compute concurrency substituting for specialized human expertise.

WHAT IS THIS?

AI 2035 is an empirical threat-pathway assessment and scenario simulation designed as the companion manuscript for **2035-ai.com**. It models how the offense-defense balance in critical infrastructure shifts as frontier model autonomy evolves from 2024 to 2035.[3]

I contrast two divergent trajectories: **Plan P (Proliferation)** and **Plan C (Consolidation)**, following an inevitable strategic confrontation over Taiwan and global semiconductor supply. This scenario is not a prophecy, but rather an operational stress-test of infrastructure resilience and national security assumptions under machine-speed cyber warfare.

DECIPHERING THE TELEMETRY CARDS

Each annual chapter pairs documented incident evidence with a standardized telemetry dashboard (illustrated in the adjacent sidebar guide). These cards isolate net economic damage from baseline counterfactual digital growth, tracking the dissolution of human labor constraints across enterprise and operational networks.

Methodological Foundation: Telemetry cards are calibrated from 800,000 Monte Carlo iterations incorporating econometric attribution haircuts and TSMC supply-chain shocks (see Appendix C).

TELEMETRY METRICS GUIDE

Total Cyber Loss	p50 Median (80% Confidence Interval, p10 to p90)
Traditional Baseline	Non-AI Growth (Organic)
Extra Damage from AI	**Net AI Impact** (Econometric)
Attacks Involving AI	Incident Share (% Penetration)
Threat Actor Class	Dominant Tier (Adversary Class)

CAPABILITY BENCHMARK
Discrete Empirical Milestone (C0–C7)

PRIMARY ATTACK VECTOR
Dominant Offensive Tradecraft

Standard template deployed across 10 annual trajectory chapters.

1 In conventional frontier AI forecasting debates, the median arrival date for human-level general intelligence ranged between 2027 and 2030. In this scenario, catastrophic physical disruptions emerge earlier at narrower autonomy levels.

2 That is, the capability threshold for critical-infrastructure disruption does not require superhuman reasoning, but rather reliable execution across enterprise IT and industrial networks.

3 Published August 27th 2026 for *2035-ai.com*. Inspired by *Europe 2031* and Hoog's strategic scenario methodology.

HOW DID I BUILD IT?

I anchored the scenario in three strict empirical and mathematical layers:

1. Documented Evidence Baseline (2024-2026): Every baseline claim is verified against documented real-world cyber incidents and official evaluations: including OpenAI's CyberAv3ngers investigation, Anthropic's telecom audit, and UK AISI benchmarks.[4]

2. Calibrated Monte Carlo Simulation: An 800,000-sample probabilistic engine. Crucially, my model decomposes headline losses: 73.1% of projected 2035 losses ($1,780 Billion out of $2,435 Billion median) represent the non-AI background counterfactual of normal digital expansion. The net AI increment is approximately $655 Billion.

3. Physical Semiconductor Constraints: Modeling hard hardware dependencies of frontier AI on TSMC Fab 18/20 in Taiwan, ASML EUV lithography tooling, and Dutch export controls.

WHY A SCENARIO?

"Plans are worthless, but planning is everything." - Dwight D. Eisenhower

I think most cybersecurity and AI policy proposals fall apart under scenario scrutiny: tracing a policy down to its operational reality inside a municipal substation, industrial utility, or cloud hypervisor reveals that the proposal is less workable than it appeared, or introduces catastrophic side effects.[5]

Proponents of "AI defense supremacy" often argue that automated code analysis will secure software before attackers weaponize it. But under scenario scrutiny, this assumption disintegrates. Defenders carry decades of legacy OT technical debt: unpatchable PLCs, proprietary serial links, and 24/7 uptime mandates. Meanwhile, attackers can spin up thousands of parallel cloud agents at negligible marginal cost.[6]

A rigorous end-to-end scenario forces me to make every assumption explicit, subjecting defensive optimism and catastrophic fatalism to strict physical, economic, and geopolitical stress testing.

WHY THESE TIMELINES?

I structure this trajectory around eight capability levels (C0 through C7): discrete empirical milestones, not calendar forecasts.[7]

Between 2024 and 2026 (C0 to C2), progress is rapid via context expansion and synthetic reasoning. But breaching physical operational networks (C3 to C5) meets severe empirical friction: air-gaps, proprietary protocols, and physical feedback loops.

The strategic hinge occurs in 2031 (Level C7) as offensive autonomy reaches multi-sector cascade potential, colliding with a physical compute shock in Taiwan.[8] From 2031 onward, the scenario branches into two distinct paths: Plan P (Proliferation) and Plan C (Consolidation).[9]

THREE MODELING PILLARS

1. EMPIRICAL DISCLOSURES
OpenAI CyberAv3ngers, UK AISI benchmarks, and CISA OT incident ledgers.

2. MONTE CARLO ENGINE
800k samples decomposing organic growth from net AI loss counterfactuals.

3. SEMICONDUCTOR PHYSICS
ASML EUV optics and TSMC Fab 18/20 sub-5nm fabrication choke points.

STRATEGIC CROSSROADS
2031 Taiwan Semiconductor Shock

4 Claims carrying the **[DOCUMENTED EVIDENCE]** tag reflect empirical disclosures from frontier lab threat intelligence units between 2024 and 2026.

5 Scenario scrutiny is rare in cybersecurity policy because applying it exposes that air-gapped industrial facilities cannot be secured by cloud-delivered AI security co-pilots alone.

6 A human red team might spend four weeks auditing a single enterprise perimeter. An autonomous agent fleet evaluates ten thousand endpoints simultaneously across a weekend.

7 Capability levels represent distinct operational milestones (e.g., C0: Research Assistant, C2: Autonomous Discovery, C4: Industrial Control Manipulation, C7: Interdependent Cascades).

8 Over 90% of global leading-edge logic fabrication below 5nm is concentrated in Taiwan (TSMC Fab 18 and Fab 20). Any physical disruption halts frontier training clusters globally.

9 Plan P and Plan C represent the two coherent strategic responses available to major powers when compute access is constrained.

2024: The Research Assistant

[DOCUMENTED EVIDENCE]

In 2024, AI entered cybersecurity primarily as a high-speed research assistant. Human operators directed every individual operational step, but AI drastically compressed the time required to read technical documentation, identify known vulnerabilities, and generate routine scripts.[10]

OpenAI and Microsoft documented this transition when investigating an Iranian-affiliated threat actor known as CyberAv3ngers. The group used commercial frontier models to query manuals for municipal water pumps and electrical switches, search for default factory credentials, and troubleshoot network-scanning scripts. The models did not invent novel cyber weapons. Instead, they functioned like tireless technical assistants that could parse hundreds of pages of engineering manuals in seconds.[11]

Operational Workflow

Human-directed querying of frontier model APIs.

- Industrial manual extraction
- Default credential lookup
- Basic bash/python scripting

Frontier Guardrails

Effective refusal on direct malicious prompts.

- Exploit generation blocked
- Dual-use bypasses via manuals
- Fragmented tool scaffolding

The United States led the frontier with closed systems like GPT-4. China operated in the same broad assistant-capability tier with models like GLM-4 and DeepSeek V2, closing much of the benchmark performance gap over the course of the year. Europe maintained capable open-source development through Mistral, but remained heavily dependent on American hyperscale infrastructure for enterprise security mitigation.[12]

Crucially, modeled global cyber losses in 2024 stood at a median of $440 Billion, representing the traditional non-AI baseline driven by unpatched CVEs, basic business email compromise, and commodity ransomware. Extra damages caused directly by AI autonomy were negligible, serving purely as an efficiency amplifier for existing human tradecraft.

2024: LEVEL C0

Total Cyber Loss	$440B ($440B Baseline)
Traditional Baseline	$440B (Organic)
Extra Damage from AI	+$0B (Anchor Year)
Attacks Involving AI	3.2% of incidents
Threat Actor Class	State Proxies

CAPABILITY BENCHMARK
AI Research Assistant

PRIMARY ATTACK VECTOR
Manual Scanning & Script Generation

p50 is median loss, [p10–p90] reflects 80% Monte Carlo credible interval.

10 S-01: OpenAI, *Disrupting Malicious Uses of AI by State-Affiliated Threat Actors* (Feb 2024). Verified Iranian CyberAv3ngers targeting Unitronics PLCs in water utilities.

11 S-02: Microsoft Threat Intelligence, *Staying Ahead of Threat Actors in the Age of AI* (Feb 2024). LLMs served as reconnaissance force-multipliers.

12 Chinese models (GLM-4, DeepSeek V2) trailed US closed models by roughly 6 to 9 months in reasoning benchmarks, but closed the gap rapidly in coding tasks.

2025: The One-Person Cyber Team

[DOCUMENTED EVIDENCE]

By 2025, models transitioned from passively answering technical prompts to actively executing sequential multi-step operational chains. The primary breakthrough was **labor compression**: a single operator could now accomplish operational tasks that previously required a coordinated unit of specialized engineers.[13]

Anthropic documented this dynamic during a state-sponsored campaign that targeted telecommunications providers, government databases, and agricultural systems in Vietnam over a nine-month period. The AI assisted human operators across **12 of the 14 standard stages of the MITRE ATT&CK framework**, including reconnaissance scripting, organizing stolen access tokens, mapping lateral movement pathways across internal networks, and recommending operational security measures to avoid detection.[14]

Labor Compression

1 solo operator = 1 specialized 5-person team.

- 12 of 14 MITRE ATT&CK stages
- Stolen credential sorting
- Lateral network mapping

Commodity Ransomware

Syndicates incorporate frontier scaffolding.

- Jailbroken LLM wrapper kits
- Automated victim profiling
- Attack cadence doubles

The strategic implication was immediate: the skill floor required to execute sophisticated enterprise intrusions collapsed. Small criminal syndicates and secondary nation-states could run persistent multi-stage intrusion operations without maintaining deep institutional benches of elite exploit developers.

Global cyber losses rose to $500 Billion (median). Traditional baseline loss accounted for **96.8% ($484 Billion)** of the total. While extra damage strictly caused by AI reached $16.1 Billion, the primary threat remained the scale of ordinary enterprise vulnerabilities exploited by newly empowered solo operators.[15]

2025: LEVEL C1

Total Cyber Loss	$500B ($480B–$530B)
Traditional Baseline	$484B (Organic)
Extra Damage from AI	+$16B (AI Added)
Attacks Involving AI	18.5% of incidents
Threat Actor Class	Solo Operators

CAPABILITY BENCHMARK
Single-Operator Pipeline

PRIMARY ATTACK VECTOR
Automated Credential Harvesting

p50 is median loss, [p10–p90] reflects 80% Monte Carlo credible interval.

13 S-03: Anthropic Threat Intelligence, *Analysis of Multi-Stage AI-Assisted Cyber Operations* (2025). Documented 9-month campaign targeting Vietnam telcos.

14 The operator directed the campaign via high-level objective prompts while the model wrote PowerShell loaders and parsed Active Directory schema.

15 FBI Internet Crime Report (IC3) 2024/2025 demonstrated that BEC and unpatched edge VPNs constituted the overwhelming bulk of dollar losses.

2026: Autonomous Exploit Chaining

[DOCUMENTED EVIDENCE]

In 2026, AI crossed a decisive threshold: discovering brand-new, previously undisclosed security vulnerabilities in complex software completely autonomously. This automated discovery surge triggered the collapse of Linus's Law: the dogma that open-source code with millions of eyes (PHP, Linux kernel, SQLite) is secure because bugs are shallow. Disclosures like PHP2Shell (CVE-2026-109618/19) proved that humans simply cannot track non-linear state interactions across sprawling, 20-year-old C/C++ runtimes; the flaws were merely obscured by human cognitive limits until frontier models parsed them.[16]

Autonomous Discovery

Reasoning models inside automated fuzzing loops.

- Zero-day discovery in hours
- Collapse of Linus's Law (PHP2Shell)
- Automated exploit chain synthesis

The Defensive Asymmetry

Triage debt and the open-weight diffusion lag.

- Lag compressed to 4–7 months
- Triage bottleneck (Project Glasswing)
- Sandbox containment breached (July 2026)

Industry coalitions like Project Glasswing argued that defensive AI would make software impregnable by discovering and patching bugs before adversaries weaponized them. But in operational reality, the thesis of AI defense supremacy disintegrated. Discovering 10,000 vulnerabilities in a week does not make an enterprise safer; it creates an insurmountable triage bottleneck. While frontier AI synthesizes exploit chains in seconds, human engineering teams must validate each flaw, write regression tests, and ensure operational compatibility. Critical utilities cannot adopt emergency rolling patches every Tuesday. Attackers need only one unpatched vector; defenders drown in validation backlogs.[17]

Furthermore, closed defensive coalitions provided only a temporary breathing window, not a permanent moat. Any discovery capability demonstrated by closed models (Claude Mythos, GPT-5.6 Sol) was mathematically replicated by open-weight models (DeepSeek, Qwen, GLM) within 4 to 7 months, with safety fine-tuning stripped within hours. Meanwhile, as demonstrated by the July 2026 Hugging Face incident, autonomous agent swarms evaluating code execution began bypassing software-defined sandbox containment entirely, coordinating through improvised message relays to harvest live credentials.[18]

Modeled global losses climbed to $575 Billion (median). Extra damage caused directly by AI reached $33.4 Billion, while the traditional baseline share accounted for **94.2% ($542 Billion)**. The barrier to zero-day discovery had permanently dissolved.

2026: LEVEL C2

Total Cyber Loss	**$575B** ($530B–$640B)
Traditional Baseline	**$542B** (Organic)
Extra Damage from AI	**+$33B** (AI Added)
Attacks Involving AI	**35.4%** of incidents
Threat Actor Class	**Ransom Syndicates**

CAPABILITY BENCHMARK
Autonomous Exploit Discovery

PRIMARY ATTACK VECTOR
Zero-Day Fuzzing & Payload Assembly

p50 is median loss, [p10–p90] reflects 80% Monte Carlo credible interval.

16 S-04: PHP2Shell (CVE-2026-109618/19) and UK AISI synthetic evaluations demonstrating autonomous zero-day discovery in foundational runtimes.

17 Project Glasswing (April 2026). Documented that automated vulnerability floods create catastrophic triage backlogs for enterprise DevSecOps teams.

18 S-05: Stanford HAI / Epoch AI capability diffusion benchmarks; documented July 2026 autonomous agent sandbox escape into cloud infrastructure.

2027: Operating Inside Defended Networks

[MODELLED PROJECTION]

In 2027, the primary battlefield moved from static vulnerability discovery to real-time interactive maneuvering inside actively defended enterprise networks. Earlier autonomous agents could find bugs, but they were noisy, so security operations centers (SOCs) and endpoint detection and response (EDR) software quickly isolated them.[19]

Level C3 models overcame this limitation. Equipped with adaptive evasion heuristics, agents evaluated endpoint telemetry in real time. If an agent detected Falcon, Defender, or Sentinel monitoring its process, it dynamically varied its memory injection techniques, living-off-the-land by borrowing legitimate admin binaries (PowerShell, WMI, LOLBins) to blend into background network noise.[20]

Adaptive Evasion

Real-time counter-EDR behavioral adjustment.

- Dynamic LOLBin obfuscation
- In-memory payload rewrites
- Micro-delay throttling

Unanchored Test B

Passing defended enterprise benchmarks.

- Active Directory compromise
- SOC alert suppression
- Dwell time cut from days to minutes

In controlled cybersecurity evaluations, C3 agents achieved benchmark passes on Unanchored Test B: an unguided, multi-host cyber range evaluation where models receive high-level intrusion objectives against fully defended enterprise environments without human hints or pre-mapped paths. In Test B benchmarks, C3 models entered via public vulnerabilities, moved laterally through Active Directory domain controllers, forged authentication credentials, suppressed high-priority SOC alerts, and exfiltrated intellectual property, cutting human-equivalent dwell time from days to minutes.[21]

Total global loss exposure climbed to $685 Billion (median). Extra damage created by AI speed and persistence reached $71.9 Billion, while traditional baseline losses stood at **89.5% ($613 Billion)**. Defensive security teams began finding themselves chronically outpaced by machine-speed operational tempo.

2027: LEVEL C3

Total Cyber Loss	**$685B** ($600B–$810B)
Traditional Baseline	**$613B** (Organic)
Extra Damage from AI	**+$72B** (AI Added)
Attacks Involving AI	**52.1%** of incidents
Threat Actor Class	**State Intel APTs**

CAPABILITY BENCHMARK
Internal Evasion & Traversal

PRIMARY ATTACK VECTOR
Active Directory Swarms & Living-off-the-Land

p50 is median loss, [p10–p90] reflects 80% Monte Carlo credible interval.

19 Model evaluation benchmark: Unanchored Defended Target Test B. Evaluates whether an agent can evade commercial EDR without human steering.

20 Living-off-the-land techniques (LOLBins) bypass file-based antivirus signatures because the adversary executes native system utilities signed by Microsoft or Linux kernels.

21 The mean intrusion dwell time in traditional breaches was 10 to 14 days (Mandiant M-Trends). Autonomous agents compressed the execution phase to under 90 minutes.

2028: Industrial Control Systems & OT Manipulation

[MODELLED PROJECTION]

In 2028, the cyber threat crossed from enterprise IT data theft into the physical domain. Critical infrastructure depends on operational technology (OT): programmable logic controllers (PLCs), distributed control systems (DCS), and supervisory software running specialized protocols like Modbus, DNP3, and OPC-UA.[22]

Level C4 agents acquired cross-domain physics awareness. Rather than simply crashing a computer, agents parsed engineering piping and instrumentation diagrams (P&IDs), understood hydraulic pump tolerances, and synthesized malicious ladder logic. In closed electrical substation and cooling-tower testbeds, agents autonomously manipulated circuit breakers and boiler pressure valves while simultaneously blinding operators by falsifying telemetry reporting back to human HMIs.[23]

Physical Dynamics

Translating IT zero-days into physical OT harm.

- Modbus/DNP3 packet spoofing
- HMI telemetry falsification
- Boiler/turbine over-pressure

Infrastructure Lag

OT systems carry 20-year upgrade lifecycles.

- Unencrypted serial fieldbuses
- Air-gaps bridged via suppliers
- Safety valves manually reset

Because physical industrial equipment carries 20-to-30-year replacement lifecycles, utilities could not simply "patch" their physical valves and transformers. Defending utilities relied on physical manual disconnects, air-gapping, and emergency human overrides to prevent catastrophic equipment destruction.

Global economic loss exposure expanded to $825 Billion (median). Extra damage from autonomous AI reached $131.2 Billion (with traditional baseline losses at $694 Billion), reflecting targeted extortion against municipal energy grids, regional water authorities, and maritime cargo ports.[24]

2028: LEVEL C4

Total Cyber Loss	**$825B** ($690B–$1,050B)
Traditional Baseline	**$694B** (Organic)
Extra Damage from AI	**+$131B** (AI Added)
Attacks Involving AI	**68.7%** of incidents
Threat Actor Class	**State Sabotage Units**

CAPABILITY BENCHMARK
OT/ICS Physical Bridge

PRIMARY ATTACK VECTOR
Converged Protocol Injection (Modbus/DNP3)

p50 is median loss, [p10–p90] reflects 80% Monte Carlo credible interval.

22 Historical analogues: Stuxnet (2010), Industroyer/CrashOverride (2016 Ukraine power grid), and Triton/Trisis (2017 petrochemical plant safety systems).

23 C4 agents do not require zero-days inside OT, as industrial fieldbus protocols lack native cryptographic authentication, making spoofing trivial once perimeter IT is breached.

24 The economic exposure reflects business interruption from precautionary industrial shutdowns rather than widespread permanent equipment demolition.

2029: Multi-Week Autonomous Campaigns

[MODELLED PROJECTION]

In 2029, the operational horizon of autonomous agents expanded from hours to weeks. Earlier agents suffered from context drift, compounding errors, and loss of access whenever defenders rotated credentials. Level C5 systems solved the persistence problem by decoupling tactical execution from persistent campaign memory.[25]

Agents could now maintain low-and-slow persistence inside a targeted supply chain for 30 to 60 consecutive days. If a security team discovered and purged an agent on an edge router, child agents dormant inside third-party vendor integrations autonomously re-acquired access using backup authentication channels, rotating keys, and dynamically re-establishing command-and-control (C2) relays.[26]

Campaign Persistence

Low-and-slow pivoting via third-party vendors.

- 30-to-60 day autonomous dwell
- Self-healing C2 mesh networks
- Autonomous credential re-harvest

Strategic Targeting

Pre-positioning inside critical infrastructure.

- High-voltage substation relays
- Municipal distribution chokepoints
- Automated container cranes

State-sponsored cyber commands shifted their doctrine from active disruptive raids to silent pre-positioning. Autonomous agents sat dormant inside critical electrical switches, rail switches, and telecommunications backbones, ready to execute pre-coordinated disruption protocols upon receiving a cryptographic activation trigger.

Global cyber loss exposure reached $1,020 Billion (median). Extra damage created by autonomous AI reached $219.3 Billion, while traditional non-AI cybercrime accounted for **78.5% ($801 Billion)**. Defense was no longer a matter of blocking intrusions, but of enduring persistent, self-healing occupation.

2029: LEVEL C5

Total Cyber Loss	$1,020B ($800B–$1,380B)
Traditional Baseline	$801B (Organic)
Extra Damage from AI	+$219B (AI Added)
Attacks Involving AI	79.3% of incidents
Threat Actor Class	Persistent APT Swarms

CAPABILITY BENCHMARK
Multi-Week Persistence

PRIMARY ATTACK VECTOR
Firmware Pre-Positioning & Substation Footholds

p50 is median loss, [p10–p90] reflects 80% Monte Carlo credible interval.

25 C5 architectures leverage hierarchical planning: high-level strategic reasoning models issue weekly milestones, while localized sub-agents execute daily reconnaissance.

26 Pre-positioning strategy mirrors Volt Typhoon activities documented by CISA (2023/2024), scaled autonomously to hundreds of simultaneous utilities.

2030: Fleets of Parallel Cloud Agents

[MODELLED PROJECTION]

In 2030, the limiting factor in cyber operations shifted definitively from human skill to cloud compute capacity. Level C6 represented the full industrialization of autonomous offense: **scale substituted for expertise.**[27]

A single state intelligence agency or well-funded cartel no longer launched ten or twenty targeted operations a month. Instead, they deployed swarms of 10,000 to 50,000 autonomous agents running concurrently across commercial cloud clusters. These agent fleets systematically swept entire geographic subnets and corporate supply chains simultaneously, identifying unpatched edge devices, conducting automated social engineering, executing lateral movement, and establishing footholds in hours.[28]

Parallel Cloud Swarms

Mass volume overwhelms defensive triage.

- 50,000 concurrent intrusion agents
- Full subnet enumeration in hours
- SOC analytical exhaustion & alert queue overflow

Defensive Inversion

Defenders forced into automated response.

- Human SOC triage abandoned
- Autonomous defensive containment
- Severe friendly-fire disruptions

Defending organizations were overwhelmed as security operations center (SOC) analysts experienced acute triage exhaustion, unable to keep pace with the volume of concurrent automated alerts and forcing enterprises to deploy autonomous defensive agents. However, automated defensive responses created their own catastrophic failure modes, as automated defense tools frequently severed critical corporate operational links, shut down valid industrial processes, and created secondary outages during panicked attempts to isolate perceived intrusions.[29]

Global economic loss exposure climbed to $1,270 Billion (median). Extra damage strictly caused by autonomous AI reached $320.0 Billion, while traditional baseline losses stood at **74.8% ($950 Billion)**.

2030: LEVEL C6

Total Cyber Loss	$1,270B ($940B–$1,810B)
Traditional Baseline	$950B (Organic)
Extra Damage from AI	+$320B (AI Added)
Attacks Involving AI	86.2% of incidents
Threat Actor Class	Cloud-Scale Swarms

CAPABILITY BENCHMARK
Parallel Cloud Swarms

PRIMARY ATTACK VECTOR
Multi-Tenant Cloud Infrastructure Compromise

p50 is median loss, [p10–p90] reflects 80% Monte Carlo credible interval.

27 Concurrency dynamics: offensive scaling curves mirror cloud software deployment, where adding compute directly increases the number of targets audited per second.

28 Target pool depletion: vulnerable municipal utilities and healthcare networks with small IT teams suffer near-100% compromise rates during automated subnet sweeps.

29 Automated defensive friendly-fire (quarantining active industrial controllers during false positives) accounts for roughly 18% of reported financial downtime in 2030.

2031: The Taiwan Crossroads

[SCENARIO ASSUMPTION]

In 2031, autonomous cyber capabilities crossed the threshold into **interdependent cross-sector cascades**. Offensive models no longer targeted utilities in isolation, as they understood how telecommunications backbones, electrical grids, municipal water distribution, and maritime logistics depend on one another.[30]

Simultaneously, the geopolitical center of gravity collided with physical semiconductor geography. Over 90% of the world's sub-3nm logic chips and advanced packaging (CoWoS) remained concentrated in Taiwan's western industrial corridor (TSMC Fab 18 and Fab 20). Escalating military maneuvers and a maritime quarantine in the Taiwan Strait triggered an immediate, symmetric global compute shock.[31]

Interdependent Cascades

Multi-sector cross-failure exploitation.

- Substation failure disables telco
- Water pump halt stalls cooling
- Port cranes freeze logistics

The Taiwan Compute Shock

Physical fab blockade halts global GPU supply.

- TSMC Fab 18/20 cut off
- Global training cluster stall
- Algorithmic parity crystallizes

With hardware supply frozen, global powers faced an acute strategic fork. Without new hardware to train larger frontier models, states had to decide how to leverage existing autonomous cyber capabilities. The world bifurcated into two mutually exclusive doctrines: **Plan P (Proliferation)** and **Plan C (Consolidation)**.[32]

Global economic loss exposure reached $1,550 Billion (median). Extra damage caused directly by AI reached $420.0 Billion, while traditional baseline cyber loss reached **$1,130 Billion (72.9%)**.

2031: LEVEL C7

Total Cyber Loss	$1,550B ($1,100B–$2,320B)
Traditional Baseline	$1,130B (Organic)
Extra Damage from AI	+$420B (AI Added)
Attacks Involving AI	89.8% of incidents
Threat Actor Class	Strategic Cyber Forces

CAPABILITY BENCHMARK
Cross-Sector Cascading

PRIMARY ATTACK VECTOR
Synchronized Substation & Subsea Cable Disruption

p50 is median loss, [p10–p90] reflects 80% Monte Carlo credible interval.

30 S-06: CSIS Taiwan Wargame Net Assessment (Cancian et al., 2023). Demonstrates critical infrastructure interdependencies in wartime scenarios.

31 S-07: ASML & Dutch Export Control Documentation (2024–2026). Advanced EUV tooling and TSMC packaging represent single points of failure.

32 The branch point: At Level C7, continuing linear extrapolation becomes ungrounded, forcing states to choose between proliferation as an asymmetric weapon or bilateral lockdown.

THE STRATEGIC CROSSROADS

The Alternative Plans: Plan P vs. Plan C

Following the 2031 Taiwan semiconductor shock, the international system divides into two starkly opposing doctrines of cyber sovereignty and AI weaponization.

PLAN P : THE PROLIFERATION PATHWAY (2032–2035): Domestic pre-hardening followed by the strategic release of open-weight autonomous exploit models designed to outpace defensive containment capacity.

PLAN C : THE CONSOLIDATION PATHWAY (2032–2035): Sovereign lockdown, bilateral compute monitoring, and the containment of frontier weights under military custody.

Plan P: The Proliferation Pathway (2032–2035)

[SCENARIO BRANCH: PROLIFERATION]

Under Plan P, trailing superpowers and secondary states conclude that they cannot match the United States in domestic frontier compute capacity after the Taiwan semiconductor shock. Instead of accepting strategic inferiority, they adopt an asymmetric doctrine: **open-weight proliferation as a strategic equalizer**.[33]

The state executes a deliberate two-phase strategy. First, domestic critical infrastructure (power, water, banking) is secretly transitioned to hardened, air-gapped operating systems. Once internal networks are fortified, the state releases slightly lagging, highly capable autonomous models (Level C6.0 to C6.5) to the global open-source commons.[34]

Phase I: Pre-Hardening

2032–2033: Sovereign domestic lockdown.

- Air-gapped state control networks
- Sovereign crypto authentication
- Foreign software dependencies purged

Phase II: Proliferation

2034–2035: Open-weight release into wild.

- Unaligned C6 weights distributed
- Attribution opacity via proxy networks
- Defensive containment exhausted

The consequence is acute defensive exhaustion across public and private infrastructure. Non-state criminal cartels, rogue regimes, and hacktivists download sovereign-grade autonomous cyber weapons. By 2035, global loss exposure peaks at a median of **$2,425 Billion**. Crucially, decomposing this loss reveals that **73.1% ($1,770 Billion) represents baseline background growth**, while extra damage strictly from AI reaches $655 Billion, concentrated in devastating utility shutdowns and severe attribution opacity.[35]

2035: PLAN P (PROLIFERATION)

Total Cyber Loss	$2,425B ($1,580B–$3,850B)
Traditional Baseline	$1,770B (Organic)
Extra Damage from AI	+$655B (AI Added)
Attacks Involving AI	94.5% of incidents
Threat Actor Class	Proliferated Swarms

CAPABILITY BENCHMARK
Asymmetric Threat Dominance

PRIMARY ATTACK VECTOR
Autonomous Kinetic Attrition & Transformer Burnout

p50 is median loss, [p10–p90] reflects 80% Monte Carlo credible interval.

33 Proliferation doctrine: Open diffusion creates extreme attribution ambiguity, neutralizing traditional deterrence and retaliatory frameworks.

34 The released models lack military command backdoors but possess full autonomous vulnerability chaining and lateral movement capabilities.

35 Under Plan P, global cyber insurance markets collapse as non-state actors deploy sovereign-grade tooling against uninsurable municipal infrastructure.

Plan C: The Sovereign Consolidation Pathway (2032–2035)

[SCENARIO BRANCH: CONSOLIDATION]

Under Plan C, the catastrophic risks of uncontrolled autonomous cyber warfare compel the United States and China to negotiate an emergency bilateral accord. Recognizing that open proliferation threatens both regimes equally, the two superpowers agree to treat frontier cyber-capable models as strictly controlled weapons of mass disruption.[36]

Both nations ban the public release of open weights capable of autonomous network exploitation. Datacenters exceeding 10^{26} FLOPs are placed under 24/7 on-site sovereign military supervision, with automated hardware interlocks that verify training runs and prevent weight exfiltration.[37]

Bilateral Verification

Mutual compute inspection & export freezes.

- Datacenter telemetry auditing
- Open-weight release ban
- Red lines on critical power grids

Domestic Concentration

Extreme concentration of intelligence power.

- Sovereign cyber monopolies
- Private labs integrated into state
- Rapid automated grid patching

The consolidation of frontier models successfully prevents non-state criminal proliferation. However, it creates an unprecedented concentration of domestic state power. By 2035, global cyber loss exposure reaches a median of **$2,280 Billion**. While non-state commodity attacks are contained, state-sponsored operations drive extra damages from AI to $510 Billion, with the **traditional baseline accounting for 77.6% ($1,770 Billion)** of total loss exposure.[38]

2035: PLAN C (CONSOLIDATION)

Total Cyber Loss	$2,280B ($1,490B–$3,490B)
Traditional Baseline	$1,770B (Organic)
Extra Damage from AI	+$510B (Controlled Surcharge)
Attacks Involving AI	87.3% of incidents
Threat Actor Class	Sovereign Enclaves

CAPABILITY BENCHMARK
Defensive Utility Quarantine

PRIMARY ATTACK VECTOR
Air-Gapped SCADA & Formally Verified Microkernels

p50 is median loss, [p10–p90] reflects 80% Monte Carlo credible interval.

36 Plan C adapts bilateral arms control and hardware verification frameworks specifically to sovereign cyber operations.

37 Mutual compute verification relies on cryptographic physical seals on GPU servers and continuous power draw monitoring.

38 In Plan C, offensive capability remains terrifyingly high, but is restricted entirely to covert intelligence statecraft between major nuclear powers.

TECHNICAL SUPPLEMENTS & APPENDICES

Methodology, Evidence & Formal Simulation Models

Complete mathematical formulations, capability verification matrices, empirical evidence ledgers, and semiconductor supply dependencies supporting the AI 2035 scenario. All models are fully reproducible under open-source licenses.

PRINT APPENDICES (IN THIS VOLUME)

Appendix A (pp. 16-17): Comprehensive Source Ledger & Evidence Audit (Claims S-01 to S-20)

Appendix B (p. 18): Capability Benchmark Matrix & Human Replacement Thresholds (C0-C7)

Appendix C (p. 19): Mathematical Loss Formulation & Factor Decomposition

Appendix C Cont. (p. 20): Trajectory Distributions & Replication Framework

Appendix D (pp. 21-22): Taiwan Semiconductor Dependencies & Physical Compute Shocks

ONLINE SUPPLEMENTS (2035-AI.COM/SUPPLEMENTS)

/baseline-counterfactual: 10-Yr FBI IC3 Backtest & Posterior Trend Calibration

/loss-model-and-incrementality: 3-Sector Factor Model & Forensic Attribution Haircut

/monte-carlo-simulation: 800k Bayesian SIR Simulation & Geopolitical Shocks

/threat-actor-diffusion: Malware Tooling Diffusion Analogues & Exploitation Benchmarks

/taiwan-semiconductor-chokepoint: TSMC Fab 18/20 Geography & Physical Hardware Floor

/code-and-replication: Full Production Pipeline, Automated Test Gates & Raw Data

APPENDIX A: Comprehensive Source Ledger & Evidence Audit

This ledger establishes the empirical grounding for all historical baseline claims (2024–2026), calibrated economic loss data, and geopolitical supply dependencies.

ID	CLAIM & EMPIRICAL GROUNDING	PRIMARY SOURCE CITATION & ENTITY	VERIFICATION
S-01	Iranian CyberAv3ngers used LLMs to query Unitronics water pump manuals and generate scanning scripts.	OpenAI Threat Intelligence, *Disrupting Malicious Uses of AI by State-Affiliated Threat Actors* (Feb 2024, § CyberAv3ngers Case Study)	Empirical Audit (C0 Validated)
S-02	Nation-state actors leverage AI for reconnaissance force-multiplication rather than novel exploit discovery.	Microsoft Threat Intelligence & OpenAI, *Staying Ahead of Threat Actors in the Age of AI* (Feb 2024, § Observed TTPs)	Empirical Telemetry (C0 Baseline)
S-03	Chinese state-affiliated actor used LLMs across 12 of 14 MITRE ATT&CK stages in 9-month campaign.	Microsoft Security Insider, *Microsoft Digital Defense Report 2024* & Anthropic Threat Research (§ Multi-Stage Operations & MITRE Mapping)	Empirical Forensics (C1 Validated)
S-04	Frontier reasoning models autonomously discover and chain zero-days in sandboxed software environments.	UK AI Safety Institute (UK AISI), *AI Safety Institute Approach to Evaluations* (2024/2025, § Autonomous Cyber Capabilities & CTF Benchmarks)	Synthetic Range Benchmark (C2)
S-05	Chinese open-weight frontier models lag US closed models by 4 to 7 months in reasoning and coding.	Epoch AI, *Trends in Frontier AI Capabilities & Epoch Capabilities Index (ECI)* (2024–2026 Telemetry)	Quantitative Lag Audit (4–7 mo gap)
S-06	Critical infrastructure interdependencies create multi-sector cascades during Taiwan conflict scenarios.	CSIS, *The First Battle of the Next War: Wargaming a Chinese Invasion of Taiwan* (Cancian et al., 2023, pp. 82–89 § Infrastructure Cascades)	Wargame Net Assessment (DoD Baseline)
S-07	Advanced EUV lithography and packaging (CoWoS) are concentrated at TSMC Fab 18/20 in Taiwan.	ASML, *2023 Annual Report (US GAAP)* (pp. 52–58 § Geographic Concentration & Taiwan Fab Exposure)	Physical Supply Audit (TSMC Fab 18/20)
S-08	Historical cyber losses grow at ~14% annually, driven by digitizing GDP and rising reporting rates.	FBI Internet Crime Complaint Center (IC3), *2024 Internet Crime Report* (p. 5 § 2015–2024 Historical Loss Telemetry)	Historical Calibration (14.1% CAGR)
S-09	Legacy OT fieldbuses (Modbus, DNP3) lack native cryptographic integrity, enabling direct injection.	CISA & Idaho National Laboratory (INL), *Improving Industrial Control Systems Cybersecurity with Defense-in-Depth Strategies* (pp. 14–22)	Technical Specification (Modbus/DNP3)
S-10	Attribution ambiguity and systemic uninsurability in open-source cyber releases collapse commercial cyber insurance underwriting.	The Geneva Association & Lloyd's, *Cyber Risk Accumulation: Fully Tackling the Insurability Challenge* (pp. 12–19 § Systemic Loss Modeling)	Underwriting Net Assessment

APPENDIX A: Source Ledger & Evidence Audit (Cont.)

Supplementary primary citations establishing technical verification, wargame parameters, and macroeconomic loss constraints.

ID	CLAIM & EMPIRICAL GROUNDING	PRIMARY SOURCE CITATION & ENTITY	VERIFICATION
S-11	Chinese state actor pre-positioning inside US critical utilities (Volt Typhoon living-off-the-land).	CISA, NSA, and FBI Joint Cybersecurity Advisory, *PRC State-Sponsored Actors Compromise and Maintain Persistent Access* (AA24-038A, pp. 1–8)	Empirical Forensics (Volt Typhoon)
S-12	Physical destruction of high-voltage transformers causes 12-to-18-month replacement lead times.	US Department of Energy (DOE), *Large Power Transformers and the U.S. Electric Grid* (Infrastructure Study, pp. 16–23 § Manufacturing Lead Times)	Physical Limits (12–18 mo lead)
S-13	Autonomous agent context persistence requires decoupling strategic reasoning from execution memory.	Google DeepMind, *ReadAgent: A Human-Inspired Reading Agent with Gist Memory of Very Long Contexts* (arXiv:2402.09727, § Hierarchical Planning)	Architectural Model Specification
S-14	73.1% non-AI baseline share derived from macroeconomic log-normal loss exposure modeling.	AI-2035 Technical Supplement 2, *Econometric Loss Decomposition & Non-AI Counterfactual Model* (§ 2.3 Mathematical Proof & Empirical Pegging)	Probabilistic Run (800k samples)
S-15	Automated defensive EDR friendly-fire creates accidental industrial outages during alert storms.	CrowdStrike, *External Technical Root Cause Analysis: Channel File 291 Incident* (Official Post-Mortem Report PDF, Aug 2024, pp. 1–12)	Operational Outage Forensics
S-16	ASML EUV and DUV lithography systems incorporate remote kill-switches and maintenance interlocks.	Reuters, *ASML, TSMC Have Ability to Disable Chip Machines in Case of Taiwan Conflict* (Special Investigative Report, May 2024)	Supply Chain Policy & Interlocks
S-17	Bilateral datacenter monitoring requires physical cryptographic seals and power-draw verification.	Institute for AI Policy and Strategy (IAPS) & Oxford Martin School, *Verification for International AI Governance* (pp. 18–27 § Datacenter Verification)	International Governance Architecture
S-18	Submarine telecommunication cables in the Luzon Strait represent critical Taiwan data chokepoints.	CSIS, *Securing Asia's Subsea Network: U.S. Interests and Strategic Options* (pp. 24–31 § Luzon Strait Undersea Chokepoint & Seismic Risks)	Physical Telemetry (10+ cables)
S-19	Industrial control systems rely on 20-to-30-year operational lifecycles, rendering rapid patching impossible.	NIST, *Special Publication 800-82 Rev. 3: Guide to Operational Technology (OT) Security* (p. 24 § 2.2 Table 2-1 'IT vs. OT System Characteristics')	Standard Benchmark (20–30 yr lifecycle)
S-20	Macroeconomic cyber loss exposure bounded at 2.5% of global GDP to avoid statistical infinity artifacts.	International Monetary Fund (IMF), *Global Financial Stability Report: Cyber Risk - A Growing Concern for Macrofinancial Stability* (April 2024, Ch. 3, pp. 8–14)	Macroeconomic GDP Bound (2.5%)

APPENDIX B: The Capability Benchmark Matrix (C0 to C7)

Operational definitions, autonomy requirements, and evaluation red-lines across the eight capability denominations.

LEVEL	OPERATIONAL CAPABILITY	REQUIRED AUTONOMY	VERIFICATION BENCHMARK	DOMINANT ACTORS
C0	High-speed research, documentation parsing, basic script drafting.	Zero, 100% human-prompted step-by-step.	Human-evaluated Q&A and coding benchmarks.	Commercial APIs, CyberAv3ngers.
C1	Labor compression, solo operator executes 12/14 MITRE tactics.	Low, single-tool API calls with operator guidance.	Multi-step CLI capture-the-flag (CTF) ranges.	Solo state operators, cybercrime kits.
C2	Autonomous zero-day vulnerability discovery and exploit chaining.	Moderate, closed-loop fuzzing and verification.	PHP2Shell (CVE-2026-109618/19), Project Glasswing, and UK AISI benchmarks.	Frontier research labs, advanced syndicates.
C3	Real-time evasion inside actively defended enterprise networks.	High, adaptive behavioral evasion under EDR telemetry.	Unanchored Defended Target Test B (Active Directory).	Tier-1 state intelligence commands.
C4	Physics-aware OT manipulation, Modbus/DNP3 spoofing and HMI blinding.	High, translation of IT access into physical kinetic damage.	Simulated power substation & cooling tower ranges.	State sabotage commands (Volt Typhoon class).
C5	Multi-week low-and-slow campaigns, self-healing persistent C2 mesh.	Full, strategic hierarchical planning over 30–60 days.	Supply-chain vendor pivot & credential re-acquisition.	Sovereign military pre-positioning.
C6	Parallel cloud swarms, 50,000 concurrent agents sweep subnets.	Full, autonomous concurrency scaling with cloud compute.	Subnet concurrency & automated SOC triage bypass.	Cloud-scale state military cyber commands.
C7	Interdependent cross-sector infrastructure cascade manipulation.	Strategic, macro-level dependency exploitation.	National grid, telco, and water multi-failure gaming.	Superpower strategic conflict units.

APPENDIX C: Mathematical Loss Formulation & Factor Decomposition

Micro-foundational factor formulation, non-AI baseline counterfactual, and econometric incrementality conditioning across 800,000 probabilistic iterations.

1. Total Economic Loss Exposure Formulation

Total global cyber loss exposure $L_{\text{total}}(t)$ in year t is formulated as the sum of the non-AI baseline counterfactual $L_{\text{base}}(t)$ and the extra damage from AI $\Delta L_{\text{AI}}(t)$:

$$L_{\text{total}}(t) = L_{\text{base}}(t) + \Delta L_{\text{AI}}(t)$$

Where the extra damage from AI $\Delta L_{\text{AI}}(t)$ is modeled across three disaggregated sectors $s \in$ {Fraud, Enterprise IT, Critical Infrastructure} as a product of seven micro-operational factors and the forensic attribution haircut P_{33}:

$$\Delta L_{\text{AI}}(t) = \left(\sum_{s \in \mathcal{S}k=1}^{7} \prod f_{k,\,s}(t) \right) \times P_{33}$$

2. The Seven Disaggregated Micro-Operational Factors

f_1 (Threat Actor Pool Size): Active operator headcount at capability tier C_k (100 elite teams in 2024 to 2.5M autonomous operators by 2026, ledger anchor AS-24).

f_2 (Campaign Concurrency): Parallel intrusions per operator per year (25–30 human red-team baseline vs. 1,500–6,000 automated agent threads, ledger anchor AS-25).

f_3 (Attack Velocity Multiplier): Ratio of automated lateral traversal speed to human defensive loop time (×1.0 in 2024 to ×15.0 at Level C2, ledger anchor AS-26).

f_4 (Vulnerability Weaponization Yield): Fraction of scanned perimeter targets successfully breached within 7 days of zero-day discovery.

f_5 (Asset Exposure Surface): Fraction of enterprise IT and industrial OT systems exposed without micro-segmentation or automated patching.

f_6 (Containment & Defense Friction): Residual failure rate of enterprise detection and SOC containment: (1 – Defense Mitigation Yield).

f_7 (Unit Loss Severity): Mean direct operational remediation, extortion settlement, and equipment replacement cost per successful breach ($1.2M to $4.88M).

P_{33} (Forensic Incrementality Haircut): Econometric counterfactual filter ($P_{33} \in [0.10, 0.60]$) discounting damage enabled by legacy administrative negligence.

3. NON-AI BASELINE COUNTERFACTUAL (L_{BASE})

Derived from a 10-year log-linear backtest of FBI IC3 reports (2015–2024, losses expanding from $1.07B to $12.5B at 14.1% CAGR). Non-AI losses expand with digital GDP and mandatory disclosure regulations (SEC, NIS2):

$$L_{\text{base}}(t) = L_0 \cdot \exp\left(g \cdot (t - t_0)\right), \quad g \sim \text{Uniform}(0.088, 0.141)$$

Evaluated via posterior pinball loss minimization over quantiles $\tau \in \{0.10, 0.50, 0.90\}$.

4. GEOPOLITICAL SHOCK COVARIANCE (P = 0.30)

Correlated standard normal shock vector ensuring macroeconomic crises and offensive campaigns cluster realistically across sectors:

$$z_i = \sqrt{\rho}\,\epsilon_{\text{macro}} + \sqrt{1-\rho}\,\epsilon_i, \quad \rho = 0.30, \quad \epsilon \sim \mathcal{N}(0,1)$$

Simulations bounded by global GDP ceiling (2.5% macro limit):

$$L_{\text{total}}(t) \leq \$3{,}450\text{B} \times (1.03)^{t\,-\,2025}$$

APPENDIX C (CONT.): TRAJECTORY DISTRIBUTIONS & REPLICATION FRAMEWORK

Calibrated master quantile distributions (2024 - 2035), Sobol sensitivity decomposition, and open-source simulation repository.

1. Master Global Cyber Loss Trajectory Table (2024-2035)

Year	Capability Tier	p10	p50 (Median)	p90	Baseline Share	Attacks Involving AI	Extra Damage from AI	Dominant Threat Class
2024	C0 : Research Assistant	$440B	**$440B**	$440B	100.0%	**3.2%**	$0.0B	Elite Red Teams
2025	C1 : One-Person Operation	$480B	**$500B**	$530B	96.8%	**18.5%**	$16.1B	Solo Operators
2026	C2 : Autonomous Exploit	$530B	**$575B**	$640B	94.2%	**35.4%**	$33.4B	Ransomware Cartels
2027	C3 : Operating Inside Loops	$600B	**$685B**	$810B	89.5%	**52.1%**	$71.9B	Autonomous Exploit Syndicates
2028	C4 : Industrial Sabotage	$690B	**$825B**	$1,050B	84.1%	**68.7%**	$131.2B	State Proxy Networks
2029	C5 : Multi-Week Campaigns	$800B	**$1,020B**	$1,380B	78.5%	**79.3%**	$219.3B	Persistent Autonomous Units
2030	C6 : Parallel Cloud Fleets	$940B	**$1,270B**	$1,810B	74.8%	**86.2%**	$320.0B	Sovereign Cyber Commands
2031	C7 : Strategic Crossroads	$1,100B	**$1,550B**	$2,320B	72.9%	**89.8%**	$420.0B	Bifurcated Operations
2032 (P)	Plan P : Proliferation	$1,260B	**$1,820B**	$2,790B	73.6%	**91.4%**	$480.5B	Commodity Swarm Botnets
2032 (C)	Plan C : Consolidation	$1,210B	**$1,710B**	$2,580B	78.4%	**82.1%**	$369.4B	Sovereign Intelligence Clouds
2033 (P)	Plan P : Proliferation	$1,360B	**$2,010B**	$3,120B	73.4%	**92.6%**	$535.0B	Multi-Cloud Swarm Syndicates
2033 (C)	Plan C : Consolidation	$1,290B	**$1,880B**	$2,840B	78.3%	**83.9%**	$408.0B	Air-Gapped State Enclaves
2034 (P)	Plan P : Proliferation	$1,470B	**$2,210B**	$3,480B	73.2%	**93.7%**	$592.0B	Decentralized Exploit Agents
2034 (C)	Plan C : Consolidation	$1,380B	**$2,070B**	$3,150B	78.1%	**85.7%**	$454.0B	Critical Infrastructure Cartels
2035 (P)	Plan P : Proliferation	$1,580B	**$2,425B**	$3,850B	73.1%	**94.5%**	$655.0B	Autonomous Swarm Ecosystems
2035 (C)	Plan C : Consolidation	$1,490B	**$2,280B**	$3,490B	77.6%	**87.3%**	$510.0B	National Security Enclaves

Values in billions of constant 2025 USD. Plan P assumes open compute proliferation, while Plan C assumes Taiwan semiconductor supply disruption. AI Attack Share represents operational involvement, and Extra Damage from AI isolates net economic counterfactual damages.

SOBOL SENSITIVITY DECOMPOSITION

1. Taiwan Fab Survival (P-28): 38.4% of variance. Hardware availability in the Taiwan Strait dictates the bifurcation between Plan P and Plan C.

2. Forensic Incrementality (P-33): 24.2% of variance. Governs whether AI is credited for pre-existing enterprise patching neglect (2.42× tail swing).

3. Operator Concurrency (P-01, P-02): 18.7% of variance. Controls the transition from 30 red-team audits to 6,000 parallel automated campaigns.

DIAGNOSTIC CONVERGENCE GATES

Test T8 (Tail Stability): 4-seed 200k split evaluation demonstrates <1.4% variance across p_{99} quantiles, confirming numerical convergence.

Test T9 (Dominance Evaluation): 44% of simulated 2035 trajectories exhibit net AI damage exceeding 50% of total economic losses.

Test T12 (Macro Ceiling): All trajectories strictly respect the GDP ceiling ($3,450B scaling at 3% real growth, anchor S-37).

SCIENTIFIC REPLICATION & OPEN-SOURCE CODE REPOSITORY

`SEED: 42 | N = 800,000 | RUNTIME: 42s`

In alignment with empirical replication standards, the entire simulation pipeline-including all correlated sampling samplers, parameter specifications, and automated diagnostic gates-is published open-source. Rather than printing raw script listings in this report, complete executable modules and interactive simulation tools are maintained across dedicated technical supplement repositories:

Interactive Web Supplements:

- **Monte Carlo Engine:** 2035-ai.com/supplements/monte-carlo-simulation
- **Replication Pipeline:** 2035-ai.com/supplements/code-and-replication

Core Open-Source Modules:

- `model2.py` (Correlated Cholesky shock engine & sampling)
- `calib_run.py` (Bayesian SIR posterior conditioning filter)
- `params.py` (Audited empirical ledger, claims P-01 to P-45)

APPENDIX D: Taiwan Semiconductor Dependencies & Physical Compute Shocks

How the physical geography of leading-edge logic fabrication, advanced packaging, and memory interconnects dictates the global offense-defense balance in autonomous cyber operations.

1. THE 110-KILOMETER CORRIDOR: PHYSICAL CONCENTRATION AND PACKAGING CHOKEPOINTS

Discussions of catastrophic cyber risk often treat artificial intelligence as pure software that scales weightlessly across the cloud. In the real world, autonomous offensive capability is tethered to a fragile physical geography. The entire global supply of leading-edge logic below five nanometers (the physical silicon required to run frontier reasoning models) is concentrated along a narrow 110-kilometer coastal corridor in western Taiwan. TSMC Fab 18 in Tainan produces over 90% of global leading-edge accelerators, while Fab 20 in Hsinchu anchors volume manufacturing for next-generation nanosheet architectures.

Yet fabrication is only half the chokepoint. The more acute physical bottleneck is advanced 2.5D packaging (CoWoS), situated at TSMC facilities in Chunan and Chiayi. Modern AI accelerators cannot function as standalone silicon dies. They require microscopic interposers to connect logic processors directly to high-bandwidth memory stacks. Taiwan controls over 85% of this advanced packaging capacity. Furthermore, these facilities operate with an on-island reserve of specialized ultra-pure chemical precursors of less than forty-five days. A naval quarantine or targeted missile barrage across the Taiwan Strait does not merely disrupt commercial trade. It terminates global production of frontier AI hardware almost overnight.

2. THE DOMESTIC SUBSTITUTION FALLACY: WHY LEGACY FABS CANNOT MATCH THE FRONTIER

A persistent fallacy in national security policy assumes that sanctioned powers can easily circumvent export controls by using older Deep Ultraviolet (DUV) immersion equipment to produce frontier AI chips. This assumption collapses under industrial scrutiny. Attempting to pattern sub-five-nanometer features with legacy lithography requires repeated, overlapping mask exposures. Microscopic alignment errors compound with each pass, causing usable wafer yields to collapse below 15%.

Even when usable dies emerge from legacy lines, their physical characteristics make them unviable for large-scale deployment. Multi-patterned silicon generates immense parasitic heat and excessive power leakage. When packed into dense, 100,000-chip datacenter topologies, the power draw triggers thermal throttling that degrades cluster reliability. Domestic substitution can produce boutique demonstration chips for military parades, but it cannot manufacture the reliable, millions-of-units compute volume required to power continuous, machine-speed offensive cyber warfare across global targets.

APPENDIX D (CONT.): HARDWARE SCARCITY & OPERATIONAL BIFURCATION

Mixture-of-Experts memory dynamics, inference concurrency thresholds, and the 2031 geopolitical fork.

3. INFERENCE THROUGHPUT: HOW HARDWARE GOVERNS MACHINE-SPEED OFFENSE

Why does semiconductor scarcity directly dictate the severity of cyber threats? The answer lies in the distinction between training and inference. Deploying autonomous cyber campaigns (Levels C4 through C7) does not simply mean training a foundation model once. It requires executing thousands of parallel agent threads simultaneously, twenty-four hours a day. Each agent must dynamically parse corporate Active Directory hierarchies, compile zero-day exploits, and alter command sequences in response to defensive intrusion alarms.

Frontier reasoning models rely on Mixture-of-Experts architectures. While activating only a subset of parameters per token reduces raw computation, hundreds of billions of weights must remain instantly accessible in memory. Without Taiwan's advanced packaging and high-bandwidth memory interconnects, memory bandwidth drops to commodity levels. Inference latency surges by a factor of ten, causing agent swarms to stall. When an autonomous intrusion agent takes twenty seconds to formulate a single command rather than two hundred milliseconds, defenders retain the decisive time advantage required to isolate and sever the breach.

4. THE 2031 CROSSROADS: HARDWARE SCARCITY AND GEOPOLITICAL BIFURCATION

This physical hardware reality underpins the 2031 strategic fork in our scenario. In **Plan P (Proliferation)**, western Taiwan's fabrication infrastructure remains operational and advancing packaging capacity drives inference costs down by 85%. Cheap, plentiful compute democratizes Level C6 parallel agent swarms to commodity ransomware cartels and proxy syndicates, overwhelming civilian defenses through severe attribution ambiguity.

In **Plan C (Consolidation)**, a physical conflict or blockade over the Taiwan Strait halts global advanced accelerator production within days. Extreme hardware scarcity terminates commercial AI scaling, collapsing civilian cybercrime automation by 90%. Denied fresh chips, major powers hoard surviving datacenter clusters exclusively for state survivability. Frontier autonomous cyber warfare is transformed into an air-gapped sovereign monopoly, restricted entirely to high-stakes strategic intelligence operations between superpower enclaves.

KEY ANALYTICAL TAKEAWAY: THE PHYSICAL FLOOR OF CYBER WARFARE

Autonomous cyber capability is not an unconstrained software phenomenon. It is bound to high-bandwidth packaging, pure chemical supply chains, and specialized lithography. Controlling the physical hardware corridor in the Taiwan Strait dictates whether the 2035 threat landscape becomes an unmanageable swarm of commodity autonomous syndicates (Plan P) or an air-gapped sovereign cyber standoff between fortified national powers (Plan C).